Daily Algebra Drill #7

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Weighted AM–GM · Vieta jumping · Power mean inequality · Infinite products via telescoping.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition (≤15 s each)

No expansion. Recognise the pattern, write the answer.

1. Factorise: $x^5 - x$.

2. Given $ab + bc + ca = 4$ and $a + b + c = 3$, find $(a + b + c)^2 - (a^2 + b^2 + c^2)$.

3. Without a calculator: $7^2 - 5^2 + 3^2 - 1^2$.

4. The roots of $x^2 - 8x + 3 = 0$ are p and q. Find $\left(p - \dfrac{1}{p}\right)\left(q - \dfrac{1}{q}\right)$.

5. Simplify: $\dfrac{(2n)!}{(2n-1)!} - \dfrac{n!}{(n-1)!}$.

6. Factorise: $27x^3 - 8$.

7. Given $t = x - \dfrac{1}{x}$, write $t^2 + 4$ in terms of x.

8. Evaluate: $\log_2 8 + \log_2 4 - \log_2 32$. (*No log rules needed: convert to powers of 2.*)

9. Given $f(x) = 2x + 1$ and $g(x) = x^2 - 1$, find $f(g(x)) - g(f(x))$.

10. Simplify: $\dfrac{a^2 - b^2}{a - b} - \dfrac{a^2 - c^2}{a - c}$.

Section B Manipulation Drills (≤50 s each)

Fractions, structure, identities. Minimum working.

1. Evaluate the telescoping product: $\displaystyle\prod_{k=2}^{n}\left(1 - \frac{1}{k}\right)$ and find its value when $n = 100$.

2. Simplify: $\dfrac{x^2 + 3x - 10}{x^2 - x - 2} \cdot \dfrac{x^2 + x - 2}{x^2 + 6x + 5}$.

3. Find the value of $a^3+b^3+c^3$ given $a+b+c=2,\ a^2+b^2+c^2=6,\ abc=-2$.

4. The expression $p(x)=x^4+4$ can be written as a product of two quadratic factors with integer coefficients. Find them. (*This is Sophie Germain: add and subtract* $4x^2$.)

5. Given $f(x)=\dfrac{2x+1}{x-1}$, find $f^{-1}(x)$ and verify $f(f^{-1}(x))=x$.

6. Prove: the product of four consecutive integers is always one less than a perfect square. (*Let the integers be* $n-1,n,n+1,n+2$ *and pair them.*)

7. Simplify: $\dfrac{1}{1+x}+\dfrac{1}{1-x}+\dfrac{2}{1+x^2}+\dfrac{4}{1+x^4}$.

8. The equation $x^3+px+q=0$ has roots α,β,γ. Express $\alpha^3+\beta^3+\gamma^3$ in terms of p and q *without using Newton's identity.* (*Each root satisfies* $r^3=-pr-q$. *Add.*)

9. Prove that if $a+b+c=0$ then $(a^2+b^2+c^2)^2=2(a^4+b^4+c^4)$.

10. Solve: $\dfrac{x+1}{x-2}+\dfrac{x-2}{x+1}=\dfrac{10}{3}$.

Section C Substitution & Structure (≤90 s each)

One clean substitution is worth ten lines of algebra.

1. Using weighted AM–GM ($\lambda a+(1-\lambda)b\geq a^\lambda b^{1-\lambda}$ for $\lambda\in(0,1)$), prove that $\dfrac{x}{3}+\dfrac{2y}{3}\geq x^{1/3}y^{2/3}$ for $x,y>0$.

2. Find all real pairs (x,y) satisfying: $x+y=1$ and $x^4+y^4=\dfrac{7}{8}$.

3. Prove: for $a,b>0$,

$$\sqrt{\frac{a^2+b^2}{2}}\ \geq\ \frac{a+b}{2}\ \geq\ \sqrt{ab}\ \geq\ \frac{2ab}{a+b}.$$

(*Prove each* $\geq$ *from right to left using a single algebraic step each.*)

4. Solve: $27^x+27^{1-x}=28$.

5. For real a,b,c with $a+b+c=1$ and $abc>0$, prove $\dfrac{1}{a}+\dfrac{1}{b}+\dfrac{1}{c}\geq 9$.

6. The sum $S=1^3+2^3+3^3+\cdots+n^3=\left(\dfrac{n(n+1)}{2}\right)^2$. Verify for $n=3$, then use it to evaluate $3^3+4^3+\cdots+10^3$ without computing each cube.

7. Find all real solutions to $|x^2-4|=|x-2|\cdot|x+2|$.

8. Given $x,y>0$ with $x+y=4$, find the maximum of x^2y.

Section D Challenge (SMC / BMO1 difficulty)

No brute force. Each question rewards one key observation.

1. **Vieta Jumping.** Suppose a and b are positive integers with $a^2 + b^2 = k(ab + 1)$ for some positive integer k.
   (a) Show $k = 4$ is achieved by $(a, b) = (2, 2)$.
   (b) Assume a solution (a, b) with $a \geq b$ exists. Consider a as a root of $x^2 - kbx + (b^2 - k) = 0$ and let a' be the other root. Show $a' = kb - a$ and $a'b = b^2 - k$. (*This is the essence of Vieta jumping: the other root is smaller.*)

2. Find all polynomials $p(x)$ with real coefficients such that $(x - 1)p(x + 1) = (x + 2)p(x)$ for all real x.

3. For positive reals a, b, c with $a + b + c = 1$, find the minimum of
$$\frac{a}{1-a} + \frac{b}{1-b} + \frac{c}{1-c}.$$

4. Evaluate:
$$\prod_{n=2}^{\infty} \frac{n^3 - 1}{n^3 + 1}.$$
(*Factor each term as a ratio of products using sum and difference of cubes. The result telescopes.*)

5. Find all integer solutions to $x^3 + y^3 = (x + y)^2$.

"Do not worry about your difficulties in mathematics. I assure you mine are still greater." — Albert Einstein

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other prep to look at today:
Daily Integral for Calculus and Logic